

Intervals of uniform Turán densities

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Abstract

We prove that the set $\Pi_{\bullet,\infty}$ of uniform Turán densities of possibly infinite families of 3-graphs contains a terminal interval: there exists $\delta > 0$ such that $[1 - \delta, 1] \subseteq \Pi_{\bullet,\infty}$. Consequently, $\Pi_{\bullet,\infty}$ has positive Lebesgue measure and Hausdorff dimension 1. The proof combines a realization principle for projective chains of finite palettes with a spectrally controlled tower of 2-lifts. Persistent Latin-orbit deletions then yield an interval of limiting palette Lagrangians, which is transported to a neighborhood of 1 by complete joins.

1 Introduction

Given $d \in [0, 1]$ and $\eta > 0$, a 3-uniform hypergraph, or simply a 3-graph, H on n vertices is called $(d, \eta, \bullet)$ -dense if

$$|H[U]| \geq d \binom{|U|}{3} - \eta n^3 \quad \text{for every } U \subseteq V(H).$$

Thus the edge-density requirement is imposed simultaneously on every vertex subset of the host graph. For a family $\mathcal{F}$ of 3-graphs, its *uniform Turán density* $\pi_{\bullet}(\mathcal{F})$ is the supremum of the numbers d such that, for every $\eta > 0$ and $n_0 \in \mathbb{N}$, there exists an $\mathcal{F}$ -free $(d, \eta, \bullet)$ -dense 3-graph on at least n_0 vertices. We write

$$\Pi_{\bullet,\infty} := \{\pi_{\bullet}(\mathcal{F}) : \mathcal{F} \text{ is a possibly infinite family of 3-graphs}\}.$$

We allow $\mathcal{F} = \emptyset$, and hence $1 \in \Pi_{\bullet,\infty}$.

This uniformly dense version of the Turán problem was initiated by Erdős and Sós [2]; see Reiher's survey [9] for an account of its subsequent development. Unlike the ordinary Turán problem, the local density condition is sensitive not merely to the total number of edges but to their distribution throughout the host hypergraph. Even the uniform Turán density of the tetrahedron $K_4^{(3)}$ remains unknown.

Palettes have emerged as the principal finite-dimensional model for uniform Turán density. Lamaison [6] proved that palettes determine the uniform Turán density of every single 3-graph. King, Sales and Schülke [4] obtained realization results for scaled 3-graph Lagrangians, and King, Piga, Sales and Schülke [3] proved that every finite-palette Lagrangian is the uniform Turán density of a finite forbidden family. Král', Kučerák, Lamaison and Tardos [5] subsequently characterized palette colorability in terms of palette homomorphisms. Most recently, Lin, Sun, Wang and Zhou [7] extended the palette characterization to arbitrary families and to the $(k - 2)$ -uniform Turán density of k -graphs for every $k \geq 3$.

In the ordinary setting, Liu and Pikhurko [8] proved that the set of Turán densities of possibly infinite families of r -graphs contains a terminal interval for every $r \geq 3$. They also showed that the uniform Turán densities of finite families of 3-graphs are dense in a non-degenerate interval. The aim of the present paper is to obtain pointwise realization of an interval in the uniformly dense setting when the forbidden family is allowed to be infinite.

Our main result is the following.

Theorem 1.1. *There exists a constant $\delta > 0$ such that*

$$[1 - \delta, 1] \subseteq \Pi_{\bullet,\infty}.$$

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The use of possibly infinite forbidden families is necessary for an interval statement: there are only countably many finite families of finite 3-graphs. On the other hand, Theorem 1.1 immediately yields the following largeness properties of the full spectrum.

Corollary 1.2. *The set $\Pi_{\bullet,\infty}$ has positive Lebesgue measure, cardinality continuum, and Hausdorff dimension 1.*

We briefly indicate the main ingredients of the proof. The first ingredient is a realization principle for projective chains of finite palettes. Suppose that

$$P_{m+1} \longrightarrow P_m$$

is such a chain and $\lambda(P_m) \downarrow x$. Weighted random pair colorings give the lower bound $\pi_{\bullet}(\mathcal{F}) \geq x$ for a suitably chosen family $\mathcal{F}$. For the reverse inequality, we diagonalize against every finite palette of Lagrangian larger than x , using the palette separation theorem and the arbitrary-family palette characterization. The resulting statement is Proposition 3.1.

It remains to construct a branching projective system whose branch limits fill an interval. We start from a large regular graph and iterate spectrally controlled 2-lifts. The signing theorem of Bilu and Linial [1] provides a uniform two-sided spectral gap for each chosen lift and its complementary lift. The root graph supports a palette for which the uniform weighting is quantitatively stable. Passing from a parent palette to a child consists of taking the full pullback and then deleting the repeated-color patterns indexed by the complementary lift. If x is a child weighting and y is its push-forward, then the palette polynomials satisfy the exact identity

$$p_{\text{child}}(x) = p_{\text{parent}}(y) - 3C_{D^-}(x),$$

where C_{D^-} is a cubic edge functional.

The analytic core is a global cubic expander inequality. It turns the spectral gap of D^- into a positive penalty for imbalance inside the lift fibers, up to an error controlled by the deviation of the parent weighting from uniformity. Latin-orbit deletions produce errors of the same quadratic type. An orthogonal decomposition across successive fibers and an exact telescoping identity make these errors summable. Consequently, the uniform weighting is the unique global maximizer of every finite palette in the tower, independently of the permitted orbit deletions.

Persistent Latin-orbit reservoirs then provide independent choices at every scale. At level r we may delete $k_r \in \{0, 1, 2, 3, 4\}$ fresh orbits. Since the number of colors doubles at each step, the corresponding costs decay as 4^{-r} . The branch limits therefore have the form

$$L_0 - \frac{d}{N_0^2} - \frac{1}{N_0^2} \sum_{r \geq 1} k_r 4^{-r}, \quad k_r \in \{0, 1, 2, 3, 4\}.$$

The redundant base-4 digit set fills the entire interval $[0, 4/3]$. Applying the projective-chain realization principle to every branch gives a non-degenerate interval $[a, b] \subseteq \Pi_{\bullet,\infty}$.

Finally, the M -fold complete join preserves palette homomorphisms and sends a palette Lagrangian x to

$$1 - \frac{1-x}{M^2}.$$

For all sufficiently large consecutive integers M , the images of $[a, b]$ overlap and converge to 1, producing the required terminal interval.

Three results are used as external inputs: the Bilu–Linial signing theorem [1], the palette separation theorem of Král', Kučerák, Lamaison and Tardos [5], and the arbitrary-family palette characterization of Lin, Sun, Wang and Zhou [7]. We state the precise forms that we use, including the relevant coordinate conventions. All other ingredients are proved in the paper.

The paper is organized as follows. Section 2 develops the palette formalism. Section 3 proves the projective-chain realization principle. Sections 4–7 construct the palette tower and establish its multiscale stability. Section 8 constructs the persistent digit orbits and obtains a non-degenerate base interval. Section 9 applies complete joins and proves Theorem 1.1. Section 10 concludes with some further remarks.

For an integer $t \geq 0$, write $[t] := \{1, \dots, t\}$, with $[0] = \emptyset$. All graphs, hypergraphs and palettes considered below are finite unless explicitly stated otherwise. We write $a_m \downarrow a$ if (a_m) is strictly decreasing and converges to a . All vector norms are Euclidean unless the operator norm is explicitly indicated. Every supremum below is taken in the ordered interval $[0, 1]$; in particular, the supremum of the empty set is 0.

2 Preliminaries

2.1 Uniform Turán density

If H is a 3-graph and $U \subseteq V(H)$, write $e_H(U)$ for the number of edges of H contained in U . A 3-graph is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$ as a subgraph. We record the quantifiers in the definition of uniform Turán density for later reference.

Definition 2.1. For $d \in [0, 1]$ and $\eta > 0$, a 3-graph H on n vertices is $(d, \eta, \bullet)$ -dense, or uniformly (d, η) -dense, if

$$e_H(U) \geq d \binom{|U|}{3} - \eta n^3$$

for every $U \subseteq V(H)$.

For a family $\mathcal{F}$ of 3-graphs, let $A(\mathcal{F})$ be the set of all $d \in [0, 1]$ with the following property: for every $\eta > 0$ and $n_0 \in \mathbb{N}$, there exists an $\mathcal{F}$ -free, uniformly (d, η) -dense 3-graph on at least n_0 vertices. The uniform Turán density of $\mathcal{F}$ is $\pi_{\bullet}(\mathcal{F}) := \sup A(\mathcal{F})$, where we use the convention $\sup \emptyset = 0$ in the ordered interval $[0, 1]$. We allow $\mathcal{F} = \emptyset$, so $\pi_{\bullet}(\emptyset) = 1$. Thus

$$\Pi_{\bullet, \infty} = \{\pi_{\bullet}(\mathcal{F}) : \mathcal{F} \text{ is a possibly infinite family of 3-graphs}\}.$$

2.2 Palettes

A palette is a pair $P = (C, T)$, where C is a finite nonempty set of colors and $T \subseteq C^3$ is a set of admissible ordered triples. A weighting of C is a vector $x = (x_c)_{c \in C}$ in the simplex

$$\Delta_C := \left\{ x \in [0, 1]^C : \sum_{c \in C} x_c = 1 \right\}.$$

The palette polynomial and palette Lagrangian are

$$p_P(x) := \sum_{(a,b,c) \in T} x_a x_b x_c, \quad \lambda(P) := \max_{x \in \Delta_C} p_P(x).$$

The unweighted density is $d(P) := |T|/|C|^3$. In particular, $d(P) = p_P(u_C)$ for the uniform weighting $u_C := |C|^{-1} \mathbf{1}$.

Let F be a 3-graph. We say that F is P -colorable if there are a linear order $<$ on $V(F)$ and a coloring $\phi : V(F) \rightarrow C$ such that $(\phi(uv), \phi(uw), \phi(vw)) \in T$ for every edge $u < v < w$ of F . A family $\mathcal{F}$ is P -colorable if at least one of its members is P -colorable; equivalently, $\mathcal{F}$ is not P -colorable precisely when no member of $\mathcal{F}$ is P -colorable.

A homomorphism $f : P \rightarrow Q$ is a map on color sets such that admissible triples of P map to admissible triples of Q . The reverse palette $\text{rev}(P)$ has the same color set, with $(a, b, c) \in \text{rev}(P)$ if and only if $(c, b, a) \in P$. An isomorphism of palettes is a bijective homomorphism whose inverse is also a homomorphism; equivalently, it is a relabeling of the color set that preserves admissible and inadmissible triples. Reversing a witnessing vertex order shows that a 3-graph is P -colorable if and only if it is $\text{rev}(P)$ -colorable. The definition also gives $d(\text{rev}(P)) = d(P)$ and $\lambda(\text{rev}(P)) = \lambda(P)$.

Lemma 2.2 (Functoriality of palette colorability). *Let P and Q be finite palettes.*

1. *If a 3-graph F is P -colorable, then every subgraph of F is P -colorable.*
2. *If $P \rightarrow Q$, then every P -colorable 3-graph is Q -colorable.*

Proof. For the first assertion, restrict the witnessing vertex order and pair coloring to the vertex set and pairs of the subgraph. For the second, compose the witnessing pair coloring with the palette homomorphism $P \rightarrow Q$. Admissibility of every edge triple is preserved by the definition of a palette homomorphism. $\square$

Lemma 2.3 (Homomorphism monotonicity). *If $P \rightarrow Q$, then $\lambda(P) \leq \lambda(Q)$.*

Proof. Let $f : C(P) \rightarrow C(Q)$ be a homomorphism and let $x \in \Delta_{C(P)}$. Push x forward by setting $y_b := \sum_{a: f(a)=b} x_a$. Then $y \in \Delta_{C(Q)}$. Every monomial occurring in $p_P(x)$ occurs in the full expansion of $p_Q(y)$ as a nonnegative summand, counted with its multiplicity even when distinct admissible triples have the same image. Consequently, $p_P(x) \leq p_Q(y) \leq \lambda(Q)$. Maximizing over x proves the claim. $\square$

We first record an elementary probabilistic construction and then quote two external structural results about palettes. The probabilistic statement is the standard weighted random-palette construction in the convention used here. This is the usual independent pair-colouring argument; compare the discussion preceding King, Sales and Schülke [4, Lemma 3.4], whose convention interchanges our second and third palette coordinates.

Theorem 2.4 (Weighted palette construction). *For every finite palette P , every $\varepsilon, \eta > 0$, and every $n_0 \in \mathbb{N}$, there exists a P -colorable 3-graph H on at least n_0 vertices that is uniformly*

$$(\max\{\lambda(P) - \varepsilon, 0\}, \eta)\text{-dense}.$$

Proof. Write $P = (C, T)$ and choose a weighting $x \in \Delta_C$ with $p_P(x) = \lambda(P)$. Let $n \geq n_0$ be sufficiently large. Independently for every pair of vertices of $[n]$, assign color c with probability x_c . Let H consist of those triples $u < v < w$ for which $(\phi(uv), \phi(uw), \phi(vw)) \in T$. The natural order and the sampled pair coloring witness that H is P -colorable.

Put $d := \max\{\lambda(P) - \varepsilon, 0\}$. For a fixed $U \subseteq [n]$, let $X_U = e_H(U)$. Every triple of U is present with probability $p_P(x) = \lambda(P)$, and hence

$$\mathbb{E}X_U = \lambda(P) \binom{|U|}{3}.$$

Changing the color assigned to a single pair affects only triples containing that pair, so it changes X_U by at most $n - 2 < n$. If $d \binom{|U|}{3} - \eta n^3 < 0$, the required lower bound is automatic. Otherwise the bad event implies

$$X_U - \mathbb{E}X_U < -\eta n^3,$$

because $d \leq \lambda(P)$. McDiarmid's bounded-differences inequality and the bound

$$\sum_{e \in \binom{[n]}{2}} n^2 = \binom{n}{2} n^2 < \frac{n^4}{2}$$

therefore give

$$\Pr \left(X_U < d \binom{|U|}{3} - \eta n^3 \right) \leq \exp \left(-\frac{2\eta^2 n^6}{n^4/2} \right) = \exp(-4\eta^2 n^2).$$

A union bound over the at most 2^n choices of U shows that the probability of any failure is at most $2^n \exp(-4\eta^2 n^2)$, which is less than 1 for all sufficiently large n . Thus some sampled coloring produces the required H . $\square$

We quote two palette-classification results. The first is the single-palette separation theorem of Král', Kučerák, Lemaître and Tardos [5, Theorem 13]. Their inverse palette $\text{inv}(Q)$ is exactly $\text{rev}(Q)$, and their ordering of the three pair colors agrees with ours.

Theorem 2.5 (Král'–Kučerák–Lemaître–Tardos). *For finite palettes P and Q , there exists a finite 3-graph that is P -colorable but not Q -colorable if and only if there is no homomorphism*

$$P \rightarrow Q \quad \text{and no homomorphism} \quad P \rightarrow \text{rev}(Q).$$

The second quoted result is the arbitrary-family palette theorem of Lin, Sun, Wang and Zhou [7, Theorem 1.4], which extends Lemaître's single-3-graph theorem [6]. In the case $k = 3$, their $(k - 2)$ -uniform density is the uniform density in Definition 2.1. For $u < v < w$, their coordinate order is $(\phi(vw), \phi(uw), \phi(uv))$, the reverse of ours. Applying their theorem to the reverse palette, and using the invariance observations above, gives the following equivalent formulation.

Theorem 2.6 (Lin–Sun–Wang–Zhou). *For every family $\mathcal{F}$ of 3-graphs,*

$$\pi_{\bullet}(\mathcal{F}) = \sup\{d(Q) : \text{no member of } \mathcal{F} \text{ is } Q\text{-colorable}\}.$$

Remark 2.7. *Theorems 2.5 and 2.6 are quoted external results. No part of their proofs is needed below; all uses are through the displayed statements.*

3 Projective limits of finite palettes

The goal of this section is to convert an analytic limit of palette Lagrangians into the uniform Turán density of a single, possibly infinite, forbidden family. The following proposition is the realization mechanism used at the end of each branch of our construction.

Proposition 3.1 (Projective-chain realization). *Let $P_1, P_2, \dots$ be finite palettes such that $P_{m+1} \rightarrow P_m$ for every $m \geq 1$. If $\lambda(P_m) \downarrow x$, then $x \in \Pi_{\bullet, \infty}$.*

Proof. If $x = 0$, let $\mathcal{F}$ consist of the one-edge 3-graph. Then every $\mathcal{F}$ -free 3-graph is edgeless, and edgeless 3-graphs of arbitrary order show that $0 \in A(\mathcal{F})$. On the other hand, fix $d > 0$, put $\eta = d/12$, and choose n_0 so large that $\binom{n}{3} > n^3/12$ whenever $n \geq n_0$. Applying the density inequality to the full vertex set shows that no edgeless 3-graph on at least n_0 vertices is uniformly (d, η) -dense. Hence $d \notin A(\mathcal{F})$, and thus $\pi_{\bullet}(\mathcal{F}) = 0$. If $x = 1$, take $\mathcal{F} = \emptyset$; complete 3-graphs show that $1 \in A(\mathcal{F})$, and hence $\pi_{\bullet}(\mathcal{F}) = 1$. We may therefore assume $0 < x < 1$. For each m , set

$$\varepsilon_m := \min \left\{ \frac{1}{m}, \frac{\lambda(P_m)}{2} \right\}.$$

Then $\varepsilon_m > 0$ and $\varepsilon_m \rightarrow 0$.

There are only countably many finite palettes up to bijective relabeling of their color sets. Choose one representative of every isomorphism class with Lagrangian larger than x , and enumerate them as $Q_1, Q_2, \dots$. For each j , choose k_j such that $\lambda(P_{k_j}) < \lambda(Q_j)$. By Lemma 2.3, there is no homomorphism from Q_j to P_{k_j} or to $\text{rev}(P_{k_j})$. Hence Theorem 2.5 gives a finite 3-graph S_j that is Q_j -colorable but not P_{k_j} -colorable.

Set $g_0 = 0$. For every m , use Theorem 2.4, with $n_0 = g_{m-1} + 1$, to choose a P_m -colorable 3-graph G_m such that $g_m := |V(G_m)| > g_{m-1}$ and G_m is uniformly $(\lambda(P_m) - \varepsilon_m, 1/m)$ -dense.

Let t_j be a positive integer with $t_j |V(S_j)| > g_{k_j-1}$, and replace every vertex v of S_j by a cluster V_v of size t_j . Include exactly the triples meeting three distinct clusters whose three original vertices form an edge of S_j . The resulting H_j is an independent blow-up of S_j and satisfies $|V(H_j)| > g_{k_j-1}$.

To see that H_j is Q_j -colorable, order the clusters according to a witnessing order of S_j , order the vertices arbitrarily within each cluster, and copy the witnessing pair color between every two distinct clusters. Color pairs inside clusters arbitrarily; no edge of H_j contains two vertices from one cluster. Conversely, if H_j were P_{k_j} -colorable, choose one vertex from every cluster. The induced subgraph is a copy of S_j , and restricting the witnessing order and pair coloring would make this copy P_{k_j} -colorable. Relabeling the copy by its clusters would then make S_j itself P_{k_j} -colorable, a contradiction.

Set $\mathcal{F} := \{H_j : j \geq 1\}$. We claim that every G_m is $\mathcal{F}$ -free. Fix j . If $k_j > m$, then the strict increase of the sequence (g_r) gives

$$|V(H_j)| > g_{k_j-1} \geq g_m = |V(G_m)|,$$

so H_j cannot embed in G_m . If $k_j \leq m$, composition of the bonding maps gives $P_m \rightarrow P_{k_j}$. By Lemma 2.2, every subgraph of the P_m -colorable graph G_m is then P_{k_j} -colorable. Since H_j is not P_{k_j} -colorable, it again cannot embed in G_m .

We now verify the lower bound with the quantifiers in Definition 2.1. Fix $0 \leq d < x$, $\eta > 0$, and $n_0 \in \mathbb{N}$. For all sufficiently large m ,

$$g_m \geq n_0, \quad \frac{1}{m} \leq \eta, \quad \lambda(P_m) - \varepsilon_m \geq d.$$

The graph G_m is then $\mathcal{F}$ -free and uniformly (d, η) -dense. Hence $d \in A(\mathcal{F})$. Since this holds for every $d < x$, we obtain $\pi_{\bullet}(\mathcal{F}) \geq x$.

Suppose that $\pi_{\bullet}(\mathcal{F}) > x$. Choose a real number t with $x < t < \pi_{\bullet}(\mathcal{F})$. By Theorem 2.6, the supremum in that theorem is larger than t , so there exists a finite palette Q such that $d(Q) > t > x$ and no member of $\mathcal{F}$ is Q -colorable. Since $d(Q) \leq \lambda(Q)$, the palette Q is isomorphic to some Q_j . Transporting the color labels through this isomorphism, the Q_j -coloring of $H_j \in \mathcal{F}$ is also a Q -coloring, a contradiction. Therefore $\pi_{\bullet}(\mathcal{F}) = x$. $\square$

4 A spectrally controlled tower of 2-lifts

We construct an infinite tower of regular graphs in which both the selected 2-lift and its complementary lift have a uniform two-sided spectral gap. This will provide the graph-theoretic skeleton of the palette construction.

Let $G = (V, E)$ be a graph. A signing is a map $s : E \rightarrow \{+1, -1\}$. Its signed adjacency matrix A_s is defined by

$$(A_s)_{uv} := \begin{cases} s(uv), & uv \in E, \\ 0, & uv \notin E. \end{cases}$$

The positive 2-lift $L_s(G)$ has vertex set $V \times \{+1, -1\}$ and edges

$$u^\sigma v^{s(uv)\sigma} \quad (uv \in E, \sigma \in \{+1, -1\}).$$

Here and throughout, u^σ denotes the vertex (u, σ) in the fiber above u . The negative lift is $L_{-s}(G)$.

Let A_+ and A_- be the adjacency matrices of the spanning subgraphs formed by the positively and negatively signed edges, respectively. Thus $A_+ = (A(G) + A_s)/2$ and $A_- = (A(G) - A_s)/2$, and

$$A(L_s(G)) = \begin{pmatrix} A_+ & A_- \\ A_- & A_+ \end{pmatrix}.$$

The subspaces $\{(v, v) : v \in \mathbb{R}^V\}$ and $\{(v, -v) : v \in \mathbb{R}^V\}$ are invariant, carrying respectively the matrices $A(G)$ and A_s . Consequently, the spectrum of $L_s(G)$ is the union of the spectrum of G and the spectrum of A_s ; for $L_{-s}(G)$ the new spectrum is the spectrum of $-A_s$.

We use the following quoted form of the Bilu–Linial signing theorem: there is an absolute constant C such that, for every integer $d \geq 2$, every graph of maximum degree at most d admits a signing satisfying

$$\|A_s\|_{\text{op}} \leq C\sqrt{d \log^3 d};$$

see [1, Theorem 3.1].

Proposition 4.1 (Expander lift tower). *There exist an integer $d \geq 160$ and a number $\rho \leq d/100$ with the following property. For every $N_\star$ there is a sequence of connected d -regular graphs*

$$D_0, D_1, D_2, \dots$$

and signings s_r of D_r such that:

- (i) $N_r := |V(D_r)| = 2^r N_0$ and $N_0 \geq N_\star$;
- (ii) $D_{r+1} = L_{s_r}(D_r)$;
- (iii) every eigenvalue of D_r apart from the simple eigenvalue d has absolute value at most ρ ;
- (iv) the complementary lift $D_{r+1}^- := L_{-s_r}(D_r)$ is connected, and every eigenvalue of D_{r+1}^- apart from the simple eigenvalue d has absolute value at most ρ .

Proof. Choose d sufficiently large that $d \geq 160$ and

$$\rho := \max\{1, C\sqrt{d \log^3 d}\} \leq \frac{d}{100}.$$

Start with K_{d+1} and repeatedly apply the Bilu–Linial signing theorem. Every 2-lift of a simple d -regular graph is again simple and d -regular. The non-trivial eigenvalues of K_{d+1} are all -1 . At every positive lift the old eigenvalues are retained while the new eigenvalues have absolute value at most ρ . After sufficiently many preliminary lifts, the resulting graph has at least $N_\star$ vertices; call it D_0 and continue the same procedure.

We verify connectedness inductively. The initial complete graph is connected. Suppose the base graph of a lift is connected, so its eigenvalue d is simple. If the positive lift were disconnected, then the eigenvalue d of its d -regular adjacency matrix would have multiplicity at least two. One copy is inherited from the base graph, so the new spectrum, namely the spectrum of A_s , would contain d , contradicting $\|A_s\|_{\text{op}} \leq \rho < d$. For the negative lift the new spectrum is the spectrum of $-A_s$, and the same contradiction applies. Thus every positive and complementary lift is connected, while the spectral-union observation above gives the asserted bounds. $\square$

For a d -regular graph G on M vertices, the spectral conclusion is equivalently

$$\left\| A(G) - \frac{d}{M} J \right\|_{\text{op}} \leq \rho,$$

where J is the all-ones matrix.

5 The palette lift and its losses

We now place a palette on the graph tower. The root palette will have a quantitatively stable uniform maximizer. Each subsequent lift admits an exact polynomial identity in which the loss is expressed by a cubic graph functional. We also introduce Latin-orbit deletions, which will encode the branching choices later in the proof.

5.1 The root palette

Let D be a simple d -regular graph on a color set V of size N . Define the palette $P(D)$ as follows:

- every ordered triple of three pairwise distinct colors is admissible;
- for every edge $uv \in E(D)$, all permutations of (u, u, v) and of (v, v, u) are admissible;
- no monochromatic triple (u, u, u) is admissible.

Its polynomial is

$$p_D(x) = 1 - 3 \sum_v x_v^2 + 2 \sum_v x_v^3 + 3 \sum_{uv \in E(D)} (x_u^2 x_v + x_v^2 x_u). \quad (5.1)$$

At the uniform weighting $u_N = N^{-1} \mathbf{1}$,

$$p_D(u_N) = 1 - \frac{3}{N} + \frac{2 + 3d}{N^2}. \quad (5.2)$$

Lemma 5.1 (Global stability of the root palette). *For every fixed d there are constants $N_{\text{root}}(d)$ and $\kappa_d > 0$ such that, whenever D is d -regular on $N \geq N_{\text{root}}(d)$ vertices,*

$$p_D(u_N) - p_D(x) \geq \kappa_d \|x - u_N\|_2^2$$

for every $x \in \Delta_V$. In particular, u_N is the unique maximizer.

Proof. For $a, b \geq 0$, we have $a^2 b + b^2 a \leq a^3 + b^3$, because the difference is $(a + b)(a - b)^2$. Since D is d -regular, Equation (5.1) gives

$$p_D(x) \leq 1 - 3 \sum_v x_v^2 + (2 + 3d) \sum_v x_v^3. \quad (5.3)$$

Equality holds at u_N .

Set

$$\theta := \frac{1}{2(2 + 3d)}, \quad \kappa_d := \frac{\theta^3}{2}, \quad N_{\text{root}}(d) := \lceil \max\{\theta^{-1}, 6\theta^{-3}\} \rceil.$$

Let $\varphi(t) = -3t^2 + (2 + 3d)t^3$. On $[0, \theta]$, we have $\varphi''(t) = -6 + 6(2 + 3d)t \leq -3$. If $\max_v x_v \leq \theta$, then $N \geq N_{\text{root}}(d)$ gives $1/N \leq \theta$. For each v , Taylor's theorem between x_v and $1/N$ therefore gives

$$\varphi(x_v) \leq \varphi(1/N) + \varphi'(1/N)(x_v - 1/N) - \frac{3}{2}(x_v - 1/N)^2.$$

Summing and using $\sum_v (x_v - 1/N) = 0$ yields

$$\sum_v \varphi(x_v) \leq N\varphi(1/N) - \frac{3}{2} \|x - u_N\|_2^2.$$

Together with Equation (5.3), this gives the desired inequality with coefficient $3/2$. Since $0 < \theta < 1$ and hence $\kappa_d = \theta^3/2 < 3/2$, this implies the stated bound.

If $\max_v x_v > \theta$, then every admissible ordered triple is non-monochromatic. Since the polynomial of all ordered triples has value 1,

$$p_D(x) \leq 1 - \sum_v x_v^3 \leq 1 - \theta^3.$$

By Equation (5.2) and the choice of $N_{\text{root}}(d)$,

$$p_D(u_N) \geq 1 - \frac{3}{N} \geq 1 - \frac{\theta^3}{2}.$$

Since $\|x - u_N\|_2^2 \leq 1$, the asserted inequality follows with coefficient $\kappa_d = \theta^3/2$. $\square$

5.2 The spectral palette lift

Suppose that a palette $P = (V(D), T)$ contains all permutations of (u, u, v) and (v, v, u) for every $uv \in E(D)$. Let s be a signing of D , and let $\pi : V(D) \times \{+1, -1\} \rightarrow V(D)$ be the natural projection. Start with the full pullback of P , whose admissible set is

$$T^{\text{pb}} := \{(\tilde{a}, \tilde{b}, \tilde{c}) : (\pi(\tilde{a}), \pi(\tilde{b}), \pi(\tilde{c})) \in T\}.$$

Thus a child triple is admissible precisely when its projection is admissible in P . For every edge $uv \in E(D)$ and every sign σ , delete all permutations of

$$(u^\sigma, u^\sigma, v^{-s(uv)\sigma}) \quad \text{and} \quad (v^{-s(uv)\sigma}, v^{-s(uv)\sigma}, u^\sigma).$$

Denote the resulting palette by $L_s(P, D)$. The projection π is a palette homomorphism $L_s(P, D) \rightarrow P$. Moreover, the distinguished repeated-color graph in the child palette is precisely $L_s(D)$.

For a graph G and a nonnegative vector x indexed by $V(G)$, let $x^{\circ 2}$ denote the coordinatewise square of x , and define the cubic edge functional

$$C_G(x) := \sum_{uv \in E(G)} x_u x_v (x_u + x_v) = \langle x^{\circ 2}, A(G)x \rangle. \quad (5.4)$$

Lemma 5.2 (Exact lift identity). *Let $P^+ = L_s(P, D)$ and let $D^- = L_{-s}(D)$. For a weighting x of the child colors, define the push-forward by $y_u = x_{u^+} + x_{u^-}$. Then*

$$p_{P^+}(x) = p_P(y) - 3C_{D^-}(x). \quad (5.5)$$

If $|V(D)| = N$, then at the uniform weighting on $2N$ child colors the loss is

$$3C_{D^-}(u_{2N}) = \frac{3d}{4N^2}. \quad (5.6)$$

Proof. The full pullback of a parent triple contributes the product of its three aggregate weights, so the full pullback polynomial is $p_P(y)$. The deleted repeated-color triples are exactly the six ordered patterns associated with every edge of the negative lift D^- . Their total polynomial is $3C_{D^-}(x)$, proving Equation (5.5). Since D^- is d -regular on $2N$ vertices, we have $C_{D^-}(u_{2N}) = d/(2N)^2$. $\square$

5.3 Latin orbits

Let C be a color set of size N . A Latin orbit is a set

$$\mathcal{O} = \{(a_i, b_i, c_i) : i \in [N]\} \subseteq C^3$$

such that each of the maps $i \mapsto a_i$, $i \mapsto b_i$, and $i \mapsto c_i$ is a bijection. Its polynomial is $R_{\mathcal{O}}(x) := \sum_{i=1}^N x_{a_i} x_{b_i} x_{c_i}$. At the uniform weighting, $R_{\mathcal{O}}(u_N) = N^{-2}$.

Lemma 5.3. *For all $a, b, c \geq 0$,*

$$abc \geq a + b + c - 2 - ((a-1)^2 + (b-1)^2 + (c-1)^2). \quad (5.7)$$

Proof. The difference between the two sides is

$$F(a, b, c) := a^2 + b^2 + c^2 + abc - 3a - 3b - 3c + 5.$$

The quadratic terms imply $F(a, b, c) \rightarrow \infty$ whenever $a^2 + b^2 + c^2 \rightarrow \infty$ in $[0, \infty)^3$ (the term abc is nonnegative). Thus F is coercive and attains a minimum on this closed orthant. On the boundary, for example when $c = 0$,

$$F(a, b, 0) = \left(a - \frac{3}{2}\right)^2 + \left(b - \frac{3}{2}\right)^2 + \frac{1}{2} > 0,$$

and the other boundary cases are symmetric. At an interior critical point,

$$2a + bc = 2b + ac = 2c + ab = 3.$$

Subtracting pairs gives

$$(a - b)(2 - c) = (b - c)(2 - a) = (c - a)(2 - b) = 0.$$

If none of a, b, c equals 2, then $a = b = c$, and the critical-point equation becomes $2a + a^2 = 3$. Its roots are 1 and -3 , so nonnegativity gives $a = b = c = 1$. If, say, $c = 2$, then the first two critical-point equations give $a + b = 3/2$, whereas the third gives $ab = -1$, which is impossible for $a, b \geq 0$. The other cases are symmetric. Thus the only interior critical point is $(1, 1, 1)$, where $F = 0$. $\square$

Lemma 5.4 (Global Latin-orbit bound). *For every Latin orbit $\mathcal{O}$ on N colors and every $x \in \Delta_C$,*

$$R_{\mathcal{O}}(x) - \frac{1}{N^2} \geq -\frac{3}{N} \|x - u_N\|_2^2. \quad (5.8)$$

Proof. Put $w_c = Nx_c$ and $H = \sum_c (w_c - 1)^2$. Apply Lemma 5.3 to every triple $(w_{a_i}, w_{b_i}, w_{c_i})$ and sum over i . Since every color occurs exactly once in each coordinate,

$$\sum_i w_{a_i} w_{b_i} w_{c_i} \geq N - 3H.$$

Dividing by N^3 and using $H = N^2 \|x - u_N\|_2^2$ proves the result. $\square$

6 A global cubic expander inequality

The next lemma is the analytic core of the construction. It gives a global lower bound for the cubic loss: imbalance inside the prescribed fibers is penalized, while the only negative term is controlled by the deviation of the pushed-forward weighting from uniformity.

Lemma 6.1 (Cubic expander inequality). *Let G be a d -regular graph on $M = 2N$ vertices, paired as v^+, v^- for $v \in [N]$. Suppose*

$$\left\| A(G) - \frac{d}{M} J \right\|_{\text{op}} \leq \rho \leq \frac{d}{100}.$$

For a probability vector x on $V(G)$, define

$$y_v = x_{v^+} + x_{v^-}, \quad z_v = x_{v^+} - x_{v^-}.$$

Then

$$3C_G(x) - \frac{3d}{M^2} \geq \frac{1}{2} \frac{d}{N} \|z\|_2^2 - \frac{27}{20} \frac{d}{N} \|y - u_N\|_2^2. \quad (6.1)$$

Proof. Normalize by setting

$$q_{v^\pm} := Mx_{v^\pm}, \quad w_v := Ny_v, \quad b_v := Nz_v.$$

Then

$$q_{v^\pm} = w_v \pm b_v, \quad \sum_v w_v = N, \quad |b_v| \leq w_v.$$

Write $H := \|w - \mathbf{1}\|_2^2$ and $B := \|b\|_2^2$. Since C_G is homogeneous of degree three, we have $C_G(x) = M^{-3}C_G(q)$, while $\|z\|_2^2 = N^{-2}B$ and $\|y - u_N\|_2^2 = N^{-2}H$. Using $M = 2N$, multiplying Equation (6.1) by $M^3/3$ shows that it is equivalent to

$$C_G(q) - dM \geq \frac{4}{3}dB - \frac{18}{5}dH. \quad (6.2)$$

We first establish a coarse bound. Put $q = \mathbf{1} + g$, so $g \perp \mathbf{1}$ and $g_i \geq -1$. The two coordinates in each fiber give

$$\|g\|_2^2 = \sum_v ((w_v + b_v - 1)^2 + (w_v - b_v - 1)^2) = 2(H + B).$$

Expanding Equation (5.4),

$$C_G(q) - dM = d\|g\|_2^2 + 2\langle g, A(G)g \rangle + \langle g^{\circ 2}, A(G)g \rangle. \quad (6.3)$$

Every coordinate of $A(G)g = A(G)q - d\mathbf{1}$ is at least $-d$, and therefore the last term is at least $-d\|g\|_2^2$. Since $g \perp \mathbf{1}$, the spectral assumption gives

$$C_G(q) - dM \geq -2\rho\|g\|_2^2 = -4\rho(H + B). \quad (6.4)$$

Suppose first that $H > \frac{2}{5}B$. Then $B < \frac{5}{2}H$, and Equation (6.4) gives

$$C_G(q) - dM \geq -\frac{14}{100}dH.$$

On the other hand,

$$\frac{4}{3}B - \frac{18}{5}H < -\frac{4}{15}H.$$

Thus Equation (6.2) holds in this case.

It remains to consider $H \leq \frac{2}{5}B$. Since $|b_v| \leq w_v$,

$$B \leq \sum_v w_v^2 = N + H,$$

and consequently $H \leq \frac{2}{3}N$. Define the set of heavy fibers

$$S := \{v : w_v > 10\}, \quad W := \sum_{v \in S} w_v.$$

For $w \geq 10$,

$$w \leq \frac{10}{81}(w-1)^2, \quad w^2 \leq \frac{100}{81}(w-1)^2.$$

Hence

$$W \leq \frac{10}{81}H, \quad \sum_{v \in S} b_v^2 \leq \frac{100}{81}H. \quad (6.5)$$

Let q' be obtained from q by setting both coordinates of each heavy fiber to zero. Since the coefficients of C_G are nonnegative, $C_G(q) \geq C_G(q')$. The mean of q' is

$$\mu = 1 - \frac{W}{N} \geq 1 - \frac{10}{81} \cdot \frac{2}{3} = \frac{223}{243}.$$

Write $q' = \mu\mathbf{1} + h$ with $h \perp \mathbf{1}$. On a non-heavy fiber we have $0 \leq q_{v\pm} = w_v \pm b_v \leq 2w_v \leq 20$, while on a heavy fiber both coordinates of q' are zero. Since $0 \leq \mu \leq 1$, it follows that $\|h\|_\infty \leq 20$. Expanding as before,

$$C_G(q') = dM\mu^3 + \mu(d\|h\|_2^2 + 2\langle h, A(G)h \rangle) + \langle h^{\circ 2}, A(G)h \rangle. \quad (6.6)$$

Since $h \perp \mathbf{1}$, we have $Jh = 0$ and hence

$$\|A(G)h\|_2 = \left\| \left(A(G) - \frac{d}{M}J \right) h \right\|_2 \leq \rho\|h\|_2.$$

Consequently,

$$|\langle h^{\circ 2}, A(G)h \rangle| \leq \|h^{\circ 2}\|_2 \|A(G)h\|_2 \leq 20\rho \|h\|_2^2.$$

It follows that the coefficient of $\|h\|_2^2$ in Equation (6.6) is at least

$$\mu(d - 2\rho) - 20\rho \geq d \left(\frac{223}{243} \cdot \frac{49}{50} - \frac{1}{5} \right) = \frac{8497}{12150}d > \frac{2}{3}d.$$

Moreover, $1 - \mu^3 \leq 3(1 - \mu) = 3W/N$, so

$$dM\mu^3 - dM \geq -6dW \geq -\frac{60}{81}dH.$$

Let $B_{\text{good}} := \sum_{v \notin S} b_v^2$. For every non-heavy fiber, the two coordinates of h contribute at least $2b_v^2$, and therefore

$$\|h\|_2^2 \geq 2B_{\text{good}} \geq 2 \left(B - \frac{100}{81}H \right).$$

Combining these estimates gives

$$C_G(q) - dM \geq \frac{4}{3}dB - \left(\frac{60}{81} + \frac{4}{3} \frac{100}{81} \right) dH = \frac{4}{3}dB - \frac{580}{243}dH.$$

Since $580/243 < 18/5$, this is at least $\frac{4}{3}dB - \frac{18}{5}dH$. This proves Equation (6.2), and rescaling proves Equation (6.1). $\square$

7 Multiscale stability of the palette tower

We now combine the one-step cubic and Latin-orbit estimates across all levels of the graph tower. The key point is that the negative parent-level errors form a summable geometric series and can therefore be absorbed by the root and fiber stability terms.

Fix the graph tower from Proposition 4.1. Let $P_0 = P(D_0)$. Inductively, suppose P_r has color set $V(D_r)$ and contains the distinguished repeated patterns associated with D_r . Form the spectral lift $\tilde{P}_{r+1} := L_{s_r}(P_r, D_r)$. Delete from $\tilde{P}_{r+1}$ the union of at most four pairwise disjoint admissible Latin orbits, each consisting entirely of triples of three pairwise distinct colors, and call the resulting palette P_{r+1} . These deletions do not remove any distinguished repeated-color pattern, so the induction may continue. The natural projection is a homomorphism $P_{r+1} \rightarrow P_r$.

For a weighting x on $V(D_n)$, let $x^{(r)}$ be its push-forward to $V(D_r)$ and put $h_r := x^{(r)} - u_{N_r}$. For each $v \in V(D_r)$ define the fiber difference $\Delta_r(v) := x_{v+}^{(r+1)} - x_{v-}^{(r+1)}$. Then

$$\|h_{r+1}\|_2^2 = \frac{1}{2}\|h_r\|_2^2 + \frac{1}{2}\|\Delta_r\|_2^2. \quad (7.1)$$

Theorem 7.1 (Global tower stability). *Choose d and ρ as in Proposition 4.1. Let κ_d be the constant from Lemma 5.1, and put $B_d := 27/20 + 12/d$. If N_0 is sufficiently large that*

$$N_0 \geq N_{\text{root}}(d) \quad \text{and} \quad N_0 \geq \frac{8B_d d}{3\kappa_d}, \quad (7.2)$$

then, for every depth n , every choice at each level of at most four pairwise disjoint Latin orbits that are admissible in the intermediate spectral-lift palette and consist of triples of pairwise distinct colors, and every weighting x on $V(D_n)$,

$$p_{P_n}(u_{N_n}) - p_{P_n}(x) \geq \frac{\kappa_d}{2}\|h_0\|_2^2 + \frac{1}{40} \sum_{r=0}^{n-1} \frac{d}{N_r} \|\Delta_r\|_2^2. \quad (7.3)$$

In particular, u_{N_n} is the unique maximizer and $\lambda(P_n) = p_{P_n}(u_{N_n})$.

Proof. For $r \geq 1$, let $\mathcal{L}_r$ be the collection of Latin orbits deleted when P_r is formed, and put $\mathcal{L}_0 = \emptyset$. Thus $|\mathcal{L}_r| \leq 4$. Because the orbits in each $\mathcal{L}_r$ are pairwise disjoint, their deletion polynomials add without multiplicity. More precisely, for $0 \leq r < n$,

$$p_{P_{r+1}}(x^{(r+1)}) = p_{P_r}(x^{(r)}) - 3C_{D_{r+1}^-}(x^{(r+1)}) - \sum_{\mathcal{O} \in \mathcal{L}_{r+1}} R_{\mathcal{O}}(x^{(r+1)}).$$

At the uniform weighting, the corresponding losses are $3d/N_{r+1}^2$ and N_{r+1}^{-2} , respectively. Subtracting the two recursions and summing over r gives the exact telescoping identity

$$\begin{aligned} p_{P_n}(u_{N_n}) - p_{P_n}(x) &= p_{P_0}(u_{N_0}) - p_{P_0}(x^{(0)}) \\ &\quad + \sum_{r=0}^{n-1} \left(3C_{D_{r+1}^-}(x^{(r+1)}) - \frac{3d}{N_{r+1}^2} \right) \\ &\quad + \sum_{r=1}^n \sum_{\mathcal{O} \in \mathcal{L}_r} \left(R_{\mathcal{O}}(x^{(r)}) - \frac{1}{N_r^2} \right). \end{aligned} \quad (7.4)$$

The root term is bounded below by $\kappa_d \|h_0\|_2^2$ by Lemma 5.1. Applying Lemma 6.1 to D_{r+1}^- , with the parent fibers as the prescribed pairing, bounds the spectral term at level r by

$$\frac{1}{2} \frac{d}{N_r} \|\Delta_r\|_2^2 - \frac{27}{20} \frac{d}{N_r} \|h_r\|_2^2.$$

By Lemma 5.4, the at most four Latin deletions at level r contribute at least

$$-\frac{12}{N_r} \|h_r\|_2^2 = -\frac{12}{d} \frac{d}{N_r} \|h_r\|_2^2.$$

Thus an interior level has total h_r -coefficient at most $B_d = 27/20 + 12/d$. At $r = 0$ there is no Latin term, and at $r = n$ there is no spectral term, so the same coefficient B_d also dominates both endpoint errors. We obtain

$$p_{P_n}(u_{N_n}) - p_{P_n}(x) \geq \kappa_d \|h_0\|_2^2 + \frac{1}{2} \sum_{r=0}^{n-1} \frac{d}{N_r} \|\Delta_r\|_2^2 - B_d \sum_{r=0}^n \frac{d}{N_r} \|h_r\|_2^2. \quad (7.5)$$

Unrolling Equation (7.1) gives

$$\|h_r\|_2^2 = 2^{-r} \|h_0\|_2^2 + \sum_{i=0}^{r-1} 2^{-(r-i)} \|\Delta_i\|_2^2.$$

Since $N_r = 2^r N_0$, the coefficient of the root deviation is bounded by

$$\sum_{r=0}^n \frac{d}{N_r} 2^{-r} \leq \frac{d}{N_0} \sum_{r=0}^{\infty} 4^{-r} = \frac{4d}{3N_0}.$$

For each fixed $i < n$, the coefficient of $\|\Delta_i\|_2^2$ is bounded by

$$\sum_{r=i+1}^n \frac{d}{N_r} 2^{-(r-i)} \leq \frac{d}{N_i} \sum_{t=1}^{\infty} 4^{-t} = \frac{1}{3} \frac{d}{N_i}.$$

Consequently,

$$\sum_{r=0}^n \frac{d}{N_r} \|h_r\|_2^2 \leq \frac{4d}{3N_0} \|h_0\|_2^2 + \frac{1}{3} \sum_{r=0}^{n-1} \frac{d}{N_r} \|\Delta_r\|_2^2. \quad (7.6)$$

Substituting this into Equation (7.5),

$$\begin{aligned} p_{P_n}(u_{N_n}) - p_{P_n}(x) &\geq \left(\kappa_d - \frac{4B_d d}{3N_0} \right) \|h_0\|_2^2 \\ &\quad + \left(\frac{1}{2} - \frac{B_d}{3} \right) \sum_{r=0}^{n-1} \frac{d}{N_r} \|\Delta_r\|_2^2. \end{aligned}$$

By Equation (7.2), the first coefficient is at least $\kappa_d/2$. Since $d \geq 160$,

$$\frac{1}{2} - \frac{B_d}{3} = \frac{1}{20} - \frac{4}{d} \geq \frac{1}{40}.$$

This proves Equation (7.3). If equality $p_{P_n}(x) = p_{P_n}(u_{N_n})$ holds, then Equation (7.3) forces $h_0 = 0$ and $\Delta_r = 0$ for every $0 \leq r < n$. Equation (7.1) then gives $h_r = 0$ for all r , and in particular $x = x^{(n)} = u_{N_n}$. $\square$

From now on, choose $N_\star$ in Proposition 4.1 sufficiently large that the resulting N_0 satisfies Equation (7.2) and

$$N_0 \geq 7 \quad \text{and} \quad N_0 > \frac{2d+2}{3}. \quad (7.7)$$

8 Persistent digit orbits

We now choose the Latin orbits in a way that permits independent choices at every level.

Relabel the vertices of D_0 by $\mathbb{Z}/N_0\mathbb{Z}$. For $j \in [4]$, define

$$R_{0,j} := \{(t, t+1, t+j+1) : t \in \mathbb{Z}/N_0\mathbb{Z}\}. \quad (8.1)$$

For each fixed j , all three coordinate maps in Equation (8.1) are translations of $\mathbb{Z}/N_0\mathbb{Z}$, and hence bijections. The three entries are pairwise distinct because their pairwise differences are 1, j , and $j+1$, all nonzero modulo N_0 for $j \in [4]$ and $N_0 \geq 7$. If an ordered triple belonged to both $R_{0,j}$ and $R_{0,j'}$, comparison of its first and third coordinates would give $j = j'$. Thus these are four pairwise disjoint Latin orbits consisting of triples of pairwise distinct colors. Hence all their triples are admissible in P_0 .

Suppose $R_{r,j}$ is a continuation reservoir at level r whose triples have pairwise distinct colors. Inductively, $R_{r,j}$ is never deleted and is therefore admissible in P_r ; consequently, its complete pullback is present in the intermediate spectral lift. This pullback decomposes into four pairwise disjoint Latin orbits

$$R_{r,j}^{\alpha,\beta} := \{(a^\sigma, b^{\alpha\sigma}, c^{\beta\sigma}) : (a, b, c) \in R_{r,j}, \sigma \in \{+1, -1\}\}, \quad (8.2)$$

where $\alpha, \beta \in \{+1, -1\}$. Each set in Equation (8.2) has $2N_r = N_{r+1}$ triples, and every coordinate map is a bijection onto the child color set: the corresponding parent coordinate is a bijection, while multiplication of the sign by a fixed α or β permutes $\{+1, -1\}$. Hence each set is a Latin orbit. Moreover, every sign triple $(\sigma_1, \sigma_2, \sigma_3)$ has the unique representation

$$\sigma = \sigma_1, \quad \alpha = \sigma_2\sigma_1, \quad \beta = \sigma_3\sigma_1,$$

so the four sets are pairwise disjoint and partition the complete pullback. Pullbacks belonging to different values of j remain disjoint after projection to the parent level. Since the parent colors are distinct, none of these triples is affected by the repeated-color deletions in the spectral lift.

For each $j \in [4]$, at level $r+1$ designate

$$O_{r+1,j} := R_{r,j}^{+,+} \quad \text{and} \quad R_{r+1,j} := R_{r,j}^{+,-}.$$

The first is the j th digit orbit at that level, while the second is retained as the continuation reservoir. Thus every level has four fresh, pairwise disjoint, admissible digit orbits consisting of triples of pairwise distinct colors, regardless of all earlier digit choices.

For a digit sequence $\mathbf{k} = (k_1, k_2, \dots) \in \{0, 1, 2, 3, 4\}^{\mathbb{N}}$, define a chain $P_n(\mathbf{k})$ as follows. On the passage from level $r-1$ to level r , after performing the spectral lift, delete precisely the first k_r digit orbits $O_{r,1}, \dots, O_{r,k_r}$. When $k_r = 0$, this means that no digit orbit is deleted at level r . The continuation reservoirs are never deleted. The orbit deletions satisfy all hypotheses of Theorem 7.1, and by construction $P_{n+1}(\mathbf{k}) \rightarrow P_n(\mathbf{k})$.

Proposition 8.1 (Exact branch values). *Let $L_0 := \lambda(P_0) = 1 - 3/N_0 + (2 + 3d)/N_0^2$. For every digit sequence $\mathbf{k}$,*

$$\lambda(P_n(\mathbf{k})) = L_0 - \sum_{r=0}^{n-1} \frac{3d}{4N_r^2} - \sum_{r=1}^n \frac{k_r}{N_r^2}. \quad (8.3)$$

Consequently,

$$\lim_{n \rightarrow \infty} \lambda(P_n(\mathbf{k})) = L_0 - \frac{d}{N_0^2} - \frac{1}{N_0^2} \sum_{r=1}^{\infty} k_r 4^{-r}. \quad (8.4)$$

Proof. By Theorem 7.1, the Lagrangian at every finite level is attained uniquely by the uniform weighting. On the passage from level $r-1$ to level r , Equation (5.6) gives the spectral-lift loss $3d/(4N_{r-1}^2)$, while each of the k_r deleted Latin orbits contributes exactly N_r^{-2} . Summing these losses proves Equation (8.3). Moreover,

$$\lambda(P_r(\mathbf{k})) - \lambda(P_{r-1}(\mathbf{k})) = -\frac{3d}{4N_{r-1}^2} - \frac{k_r}{N_r^2} < 0,$$

so the finite-level Lagrangians are strictly decreasing. Since $N_r = 2^r N_0$,

$$\sum_{r=0}^{\infty} \frac{3d}{4N_r^2} = \frac{d}{N_0^2},$$

and Equation (8.4) follows. $\square$

Lemma 8.2 (Interval coding).

$$\left\{ \sum_{r=1}^{\infty} k_r 4^{-r} : k_r \in \{0, 1, 2, 3, 4\} \right\} = \left[0, \frac{4}{3} \right].$$

Proof. Because $0 \leq k_r \leq 4$, the geometric-series bound $\sum_{r \geq 1} k_r 4^{-r} \leq 4/3$ gives the inclusion from left to right. For the reverse inclusion, put $I = [0, 4/3]$. The five intervals

$$\frac{k+I}{4} = \left[\frac{k}{4}, \frac{k}{4} + \frac{1}{3} \right], \quad k = 0, 1, 2, 3, 4,$$

cover I , since consecutive intervals overlap. Given $x_0 \in I$, choose recursively $k_r \in \{0, \dots, 4\}$ and $x_r \in I$ such that $x_{r-1} = (k_r + x_r)/4$. Then, for every m ,

$$x_0 = \sum_{r=1}^m k_r 4^{-r} + 4^{-m} x_m.$$

Since $0 \leq x_m \leq 4/3$, the final term tends to zero, and the desired digit expansion follows. $\square$

Theorem 8.3 (A base interval). *With d and N_0 chosen as above, define*

$$a := L_0 - \frac{d+4/3}{N_0^2}, \quad b := L_0 - \frac{d}{N_0^2}. \quad (8.5)$$

Then $0 < a < b < 1$, and

$$[a, b] \subseteq \Pi_{\bullet, \infty}.$$

Proof. Using Equation (5.2),

$$a = 1 - \frac{3}{N_0} + \frac{2d+2/3}{N_0^2}, \quad b = 1 - \frac{3}{N_0} + \frac{2d+2}{N_0^2}.$$

Thus $a > 0$ because $N_0 \geq 7$, $b - a = 4/(3N_0^2) > 0$, and $b < 1$ by Equation (7.7).

By Proposition 8.1 and Lemma 8.2, every $x \in [a, b]$ is the limit of the strictly decreasing Lagrangian sequence along some projective palette chain $P_n(\mathbf{k})$. Reindexing this chain from 1 and applying Proposition 3.1 proves the claim. $\square$

9 Complete joins and the terminal interval

Let $P = (C, T)$ be a palette and let $M \geq 2$. Define its M -fold complete join $J_M(P)$ on the color set $[M] \times C$ as follows. A triple $((i, a), (j, b), (k, c))$ is admissible if either the macro-colors i, j, k are not all equal, or $i = j = k$ and $(a, b, c) \in T$.

Lemma 9.1 (Join formula). *For every finite palette P and integer $M \geq 2$,*

$$\lambda(J_M(P)) = 1 - \frac{1 - \lambda(P)}{M^2}. \quad (9.1)$$

Moreover, every homomorphism $P \rightarrow Q$ induces a homomorphism $J_M(P) \rightarrow J_M(Q)$.

Proof. For a weighting on $[M] \times C$, let s_i be the total weight in macro-class i . The polynomial of all ordered triples has value 1. Those whose three macro-colors are all equal have total weight $\sum_i s_i^3$; hence the contribution from triples whose macro-colors are not all equal is $1 - \sum_{i=1}^M s_i^3$. Within macro-class i , the contribution is at most $\lambda(P)s_i^3$, with equality obtainable independently in every class with $s_i > 0$; classes of zero weight contribute nothing. Hence

$$\lambda(J_M(P)) = \max_{s \in \Delta_M} \left(1 - (1 - \lambda(P)) \sum_{i=1}^M s_i^3 \right).$$

For every weighting x ,

$$0 \leq p_P(x) \leq \sum_{(a,b,c) \in C^3} x_a x_b x_c = 1,$$

so $0 \leq \lambda(P) \leq 1$. Therefore the displayed expression is maximized by minimizing $\sum_i s_i^3$. Since $\sum_i s_i^3 \geq M^{-2}$, with equality at the uniform macro-weighting, Equation (9.1) follows. The homomorphism statement is immediate by applying the original color map within each macro-class. $\square$

Proof of Theorem 1.1. Let $[a, b] \subseteq \Pi_{\bullet, \infty}$ be the interval from Theorem 8.3. Fix $M \geq 2$ and $x \in [a, b]$, and apply J_M to every palette in a projective chain realizing x . Lemma 9.1 shows that the joined Lagrangians decrease to $1 - (1 - x)/M^2$, and the joined bonding maps are homomorphisms. By Proposition 3.1,

$$\left[1 - \frac{1-a}{M^2}, 1 - \frac{1-b}{M^2} \right] \subseteq \Pi_{\bullet, \infty}.$$

Put $A = 1 - a$ and $B = 1 - b$, so $A > B > 0$. Since $((M+1)/M)^2 \rightarrow 1 < A/B$, there exists an integer $M_0 \geq 2$ such that

$$\frac{B}{M^2} \leq \frac{A}{(M+1)^2}$$

for every $M \geq M_0$. Equivalently, the left endpoint of the $(M+1)$ st joined interval is at most the right endpoint of the M th joined interval. Thus the intervals for consecutive values $M \geq M_0$ overlap, and their union is

$$\left[1 - \frac{A}{M_0^2}, 1 \right).$$

Since $1 = \pi_{\bullet}(\emptyset)$, the endpoint 1 also belongs to $\Pi_{\bullet, \infty}$. Taking $\delta := (1 - a)/M_0^2 > 0$ completes the proof. $\square$

10 Concluding remarks

The proof gives no useful numerical value of δ . The first source of quantitative loss is the Bilu–Linial degree threshold needed to make the signed spectral radius a small fraction of d ; the second is the root stability constant κ_d , which forces N_0 to be very large. Both are irrelevant to the qualitative interval statement.

The four digit orbits were chosen with quantitative slack rather than optimality in mind. A level- r digit has weight $N_r^{-2} = 4^{-r} N_0^{-2}$. Three orbit streams, giving digits 0, 1, 2, 3, would already yield the ordinary

base-4 interval $[0, 1]$; the fourth stream is not needed for interval formation, but provides overlap between adjacent digit cylinders. Other branching factors or lift operations may lead to more economical codings.

The argument is specific to the standard uniform density of 3-graphs because it uses pair-color palettes and a cubic graph functional. The palette theorem of Lin, Sun, Wang and Zhou applies to the $(k - 2)$ -uniform Turán density of k -graphs for every $k \geq 3$ [7]. This suggests that higher-uniformity analogues may be approachable through higher-dimensional lift gadgets and suitable multilinear expander inequalities.

Problem 10.1. *For every $k \geq 4$, does the set of $(k - 2)$ -uniform Turán densities of possibly infinite families of k -graphs contain a non-degenerate interval, or even a terminal interval?*

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